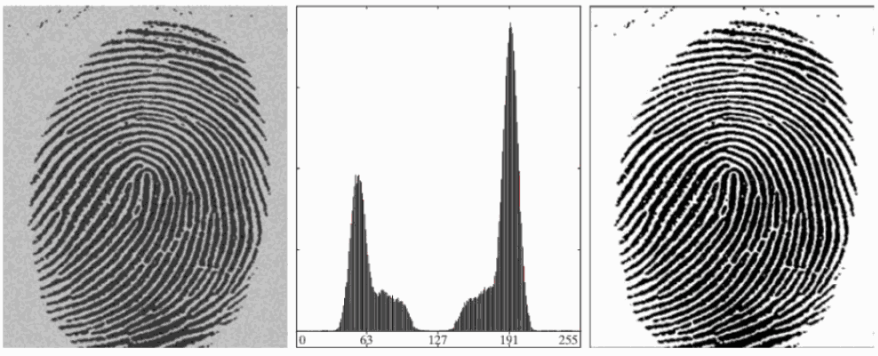
Otomatik eşikleme



a b c

FIGURE 10.38 (a) Noisy fingerprint. (b) Histogram. (c) Segmented result using a global threshold (the border was added for clarity). (Original courtesy of the National Institute of Standards and Technology.)

Tüm görüntüye aynı eşikse global eşikleme

bölgesel farklı eşikler local eşikler

Global eşikleme

$$g(x,y) = \begin{cases} 1 & \text{if } f(x,y) > T \\ 0 & \text{O.W} \end{cases}$$

Basic Global Tresholding

A) Initialize $T_0 = \frac{1}{MN} \sum_x \sum_y f(x,y)$, $i = 0, \epsilon$

B) $i \leftarrow i+1$

C) Segment the image

$(x,y) \in G_1$ if $f(x,y) \leq T_{i-1}$

$(x,y) \in G_2$ if $f(x,y) > T_{i-1}$

D) Compute μ_1, μ_2

$$\mu_1 = \frac{1}{k_1} \sum_x \sum_y f(x,y)$$

k_1: piksel sayısı

$x, y \in G_1$

$$\mu_2 = \frac{1}{k_2} \sum_x \sum_y f(x,y)$$

$x, y \in G_2$

$k_1 + k_2 = M \cdot N$

E) compute new Treshold $T_i = \frac{1}{2}(\mu_1 + \mu_2)$

F) if $|T_i - T_{i-1}| > \epsilon$ goto B

G) $T = T_i$

a b
c d

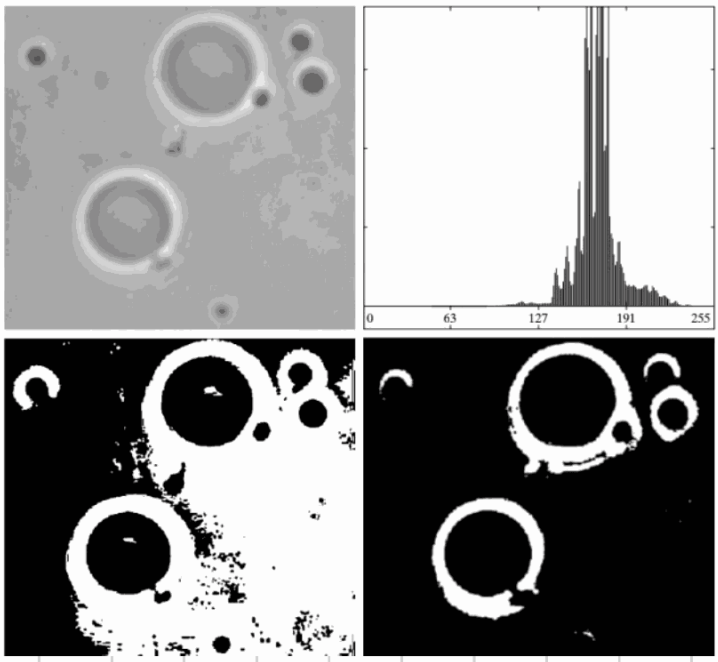
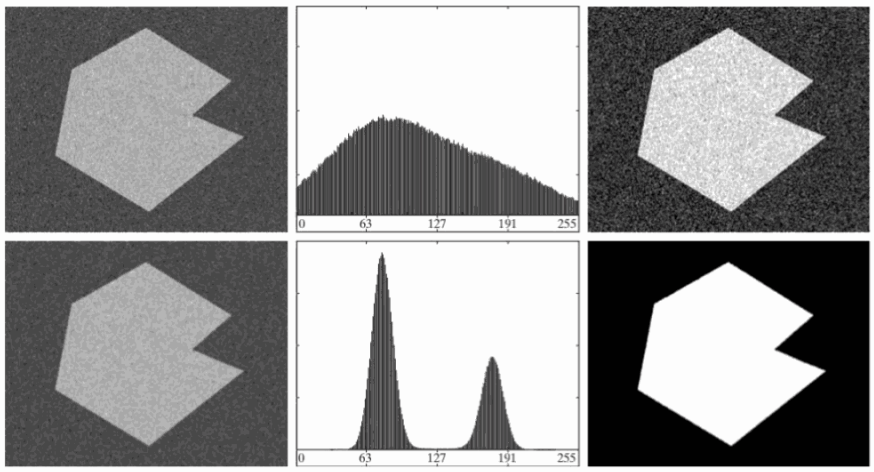
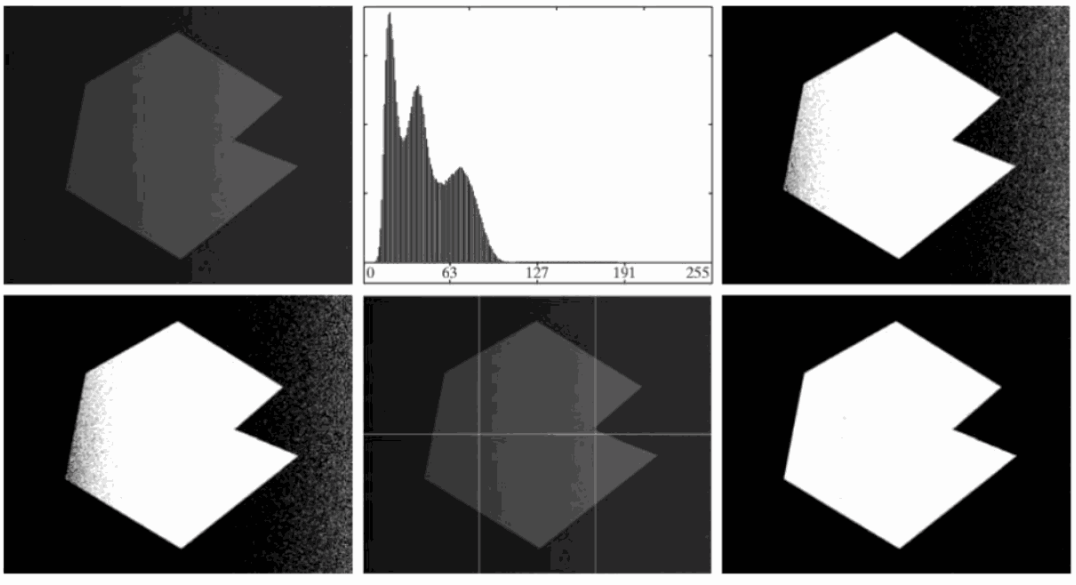


FIGURE 10.39
(a) Original image.
(b) Histogram (high peaks were clipped to highlight details in the lower values).
(c) Segmentation result using the basic global algorithm from Section 10.3.2.
(d) Result obtained using Otsu's method.
(Original image courtesy of Professor Daniel A. Hammer, the University of Pennsylvania.)



a b c
d e f

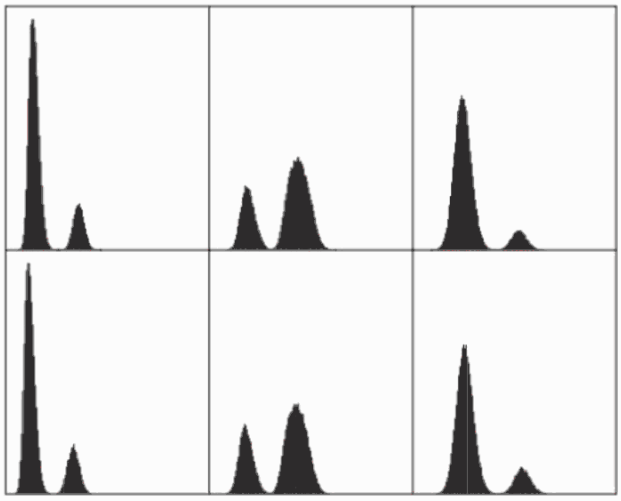
FIGURE 10.40 (a) Noisy image from Fig. 10.36 and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5×5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method.



a b c
d e f

FIGURE 10.46 (a) Noisy, shaded image and (b) its histogram. (c) Segmentation of (a) using the iterative global algorithm from Section 10.3.2. (d) Result obtained using Otsu's method. (e) Image subdivided into six subimages. (f) Result of applying Otsu's method to each subimage individually.

FIGURE 10.47
Histograms of the six subimages in Fig. 10.46(e).

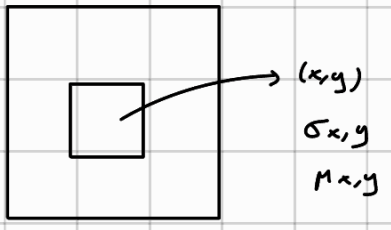




a b c

FIGURE 10.49 (a) Text image corrupted by spot shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages.

Variable Tresholding based on local image processing

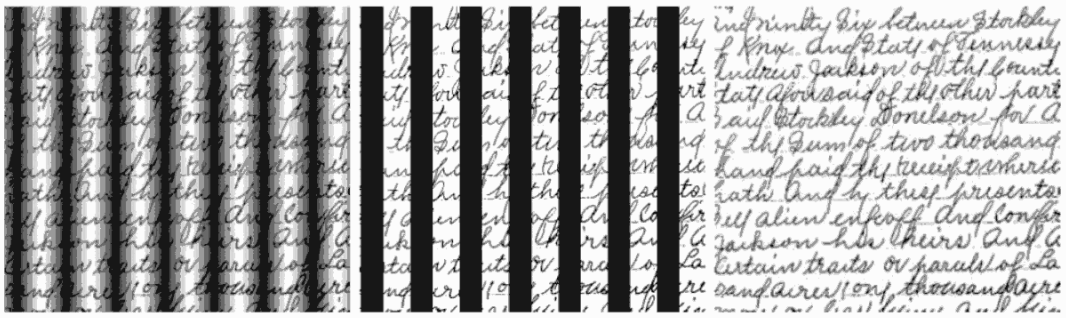


Set Treshold $T_{x,y} = a\,\sigma_{x,y} + m_{x,y}$

a>0

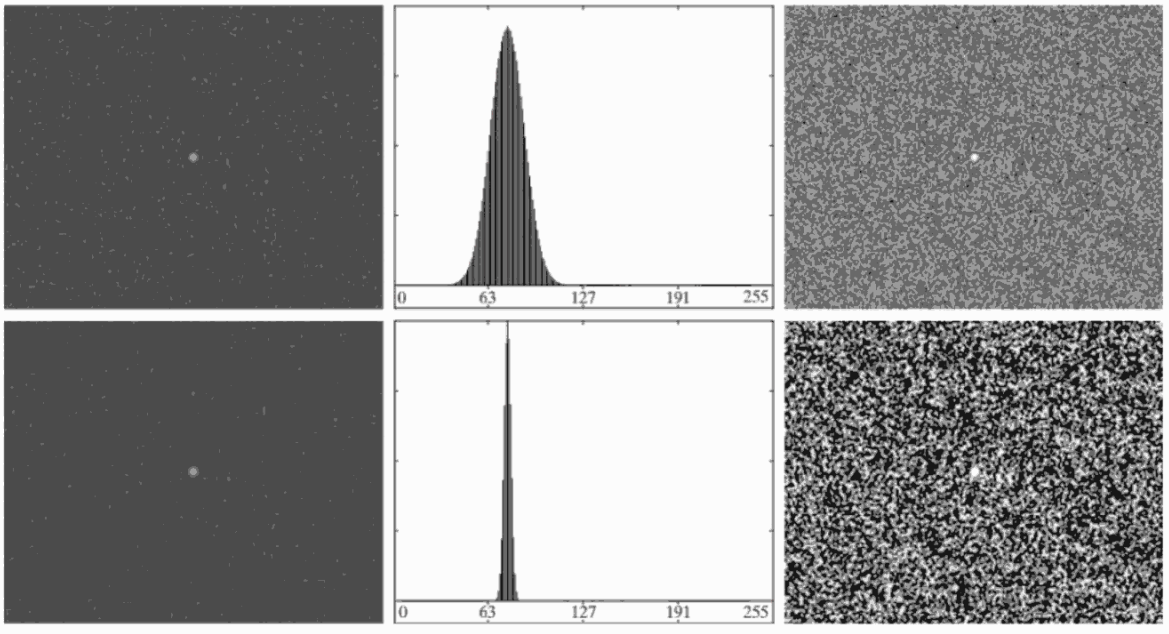


Impulse noise tuzlu biber gürültüsü



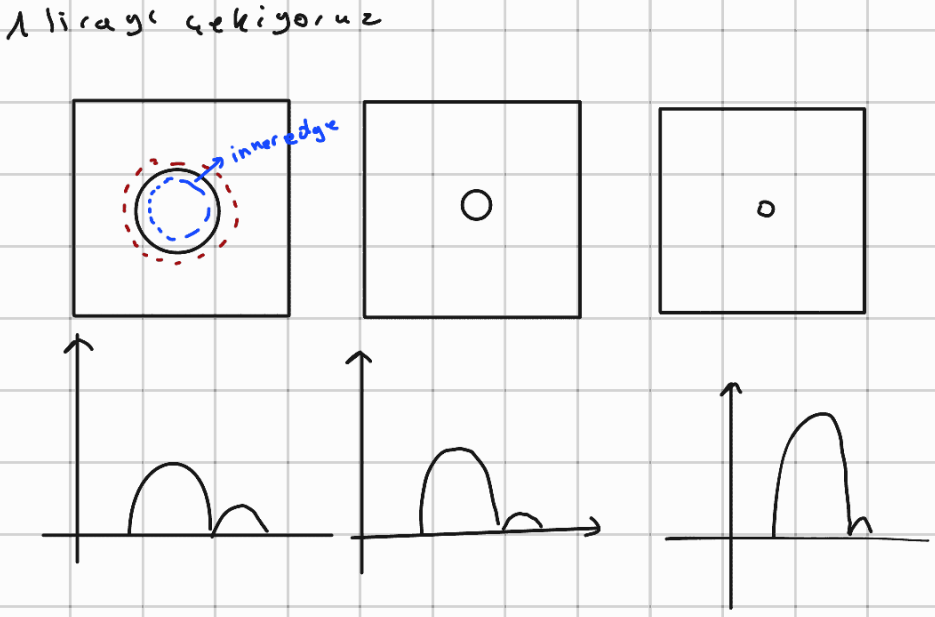
a b c

FIGURE 10.50 (a) Text image corrupted by sinusoidal shading. (b) Result of global thresholding using Otsu's method. (c) Result of local thresholding using moving averages.

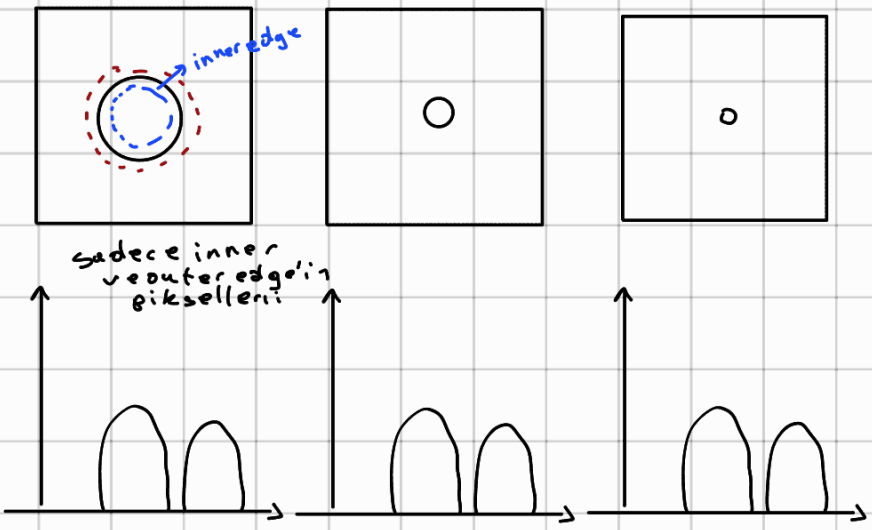


a b c
d e f

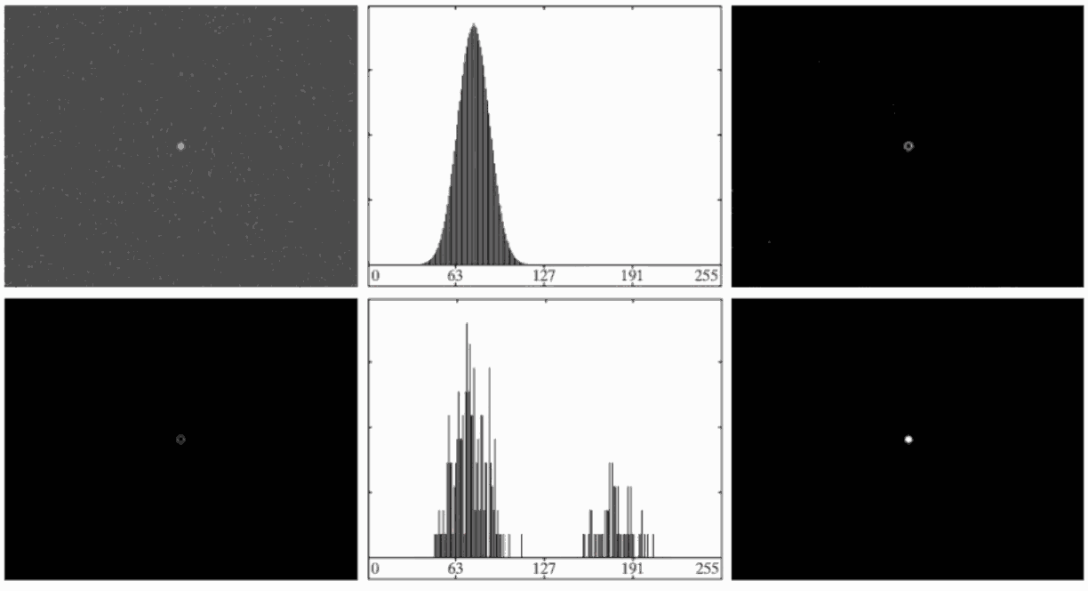
FIGURE 10.41 (a) Noisy image and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5×5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method. Thresholding failed in both cases.



Histogram hesabı yaparken sadece edge'lerden alacağız



thresholding.

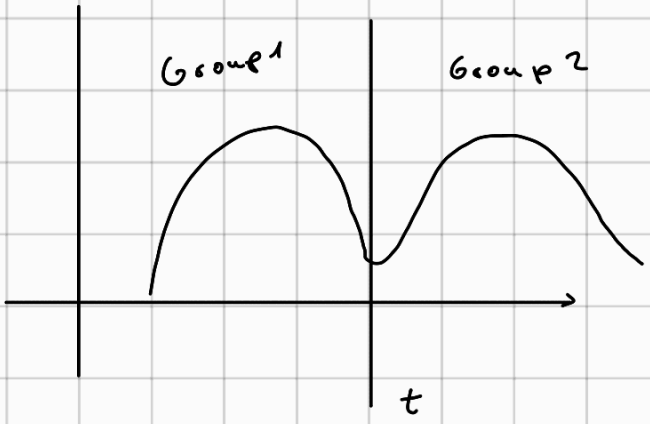


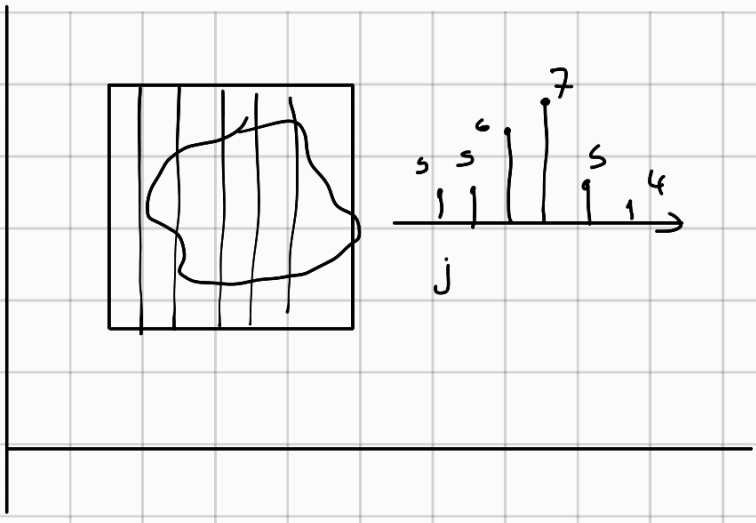
a b c
d e f

FIGURE 10.42 (a) Noisy image from Fig. 10.41(a) and (b) its histogram. (c) Gradient magnitude image thresholded at the 99.7 percentile. (d) Image formed as the product of (a) and (c). (e) Histogram of the nonzero pixels in the image in (d). (f) Result of segmenting image (a) with the Otsu threshold based on the histogram in (e). The threshold was 134, which is approximately midway between the peaks in this histogram.

Check görüntüsü var 2. baskıdan //

Otsu Tresholding





Normalized histogram $p(i)$ $i = 1, \dots, I$

I is number of graylevels

$$\sum_{i=1}^{I} p(i) = 1$$

$\mu_k(t)$ $k = 1, 2$

$\sigma_k^2(t)$

$$\mu_1(t) = \sum_{i=1}^{t} i \frac{p(i)}{q_1(t)} \qquad q_1(t) = \sum_{i=1}^{t} p(i)$$

↗ p lerin yüzde kaçı t'ninaltında

$$\mu_2(t) = \sum_{i=t+1}^{I} i \cdot \frac{p(i)}{q_2(t)} \qquad q_2(t) = \sum_{i=t+1}^{I} p(i)$$

$$q_1(t) + q_2(t) = 1$$

$$\sigma_1^2(t) = \sum_{i=1}^{t} (i - \mu_1(t))^2 \frac{p(i)}{q_1(t)}$$

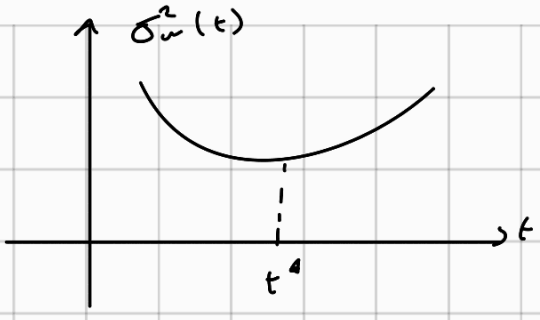
$$\sigma_2^2(t) = \sum_{i=t+1}^{I} (i - \mu_2(t))^2 \frac{p(i)}{q_2(t)}$$

within group variance

$$\sigma_w^2(t) = q_1(t)\,\sigma_1^2(t) + q_2(t)\,\sigma_2^2(t)$$

↗ ortalama variance hesabı

Otsu : minimize $\sigma_w^2(t)$



görüntünün Varyansı

$$\sigma^2 = \sum_{i=1}^{I} (i-\mu)^2 p(i)$$

$$\mu = \sum_{i=1}^{I} i\, p(i)$$

$$\underbrace{\sigma^2}_{\text{constant}} = \sigma_w^2(t) + \underbrace{q_1(t)\left[\mu_1(t)-\mu\right]^2 + q_2(t)\left[\mu_2(t)-\mu\right]^2}_{\sigma_B^2(t) \;\rightarrow \text{between group variance}}$$

maximize

Otsu:

$$\mu = \sum i\, p(t)$$

$q_1(0) = 0$

$\mu_1(0) = 0$

find $\underset{t=1,\dots,I}{\arg\max}\ \sigma_B^2(t)$

$$q_1(t) = q_1(t-1) + p(t)$$

$$q_2(t) = 1 - q_1(t)$$

$$\mu_1(t) = \left[q_1(t-1)\,\mu_1(t-1) + t\,p(t)\right]/q_1(t)$$

$$\mu_2(t) = \left[\mu - q_1(t)\mu_1(t)\right]/q_2(t)$$

$$\sigma_B^2(t) = q_1(t)q_2(t)\left[\mu_1(t)-\mu_2(t)\right]^2$$

Canny edge detection

1. Smooth the image:

$$g(x,y) = G_\sigma(x,y) * f(x,y)$$

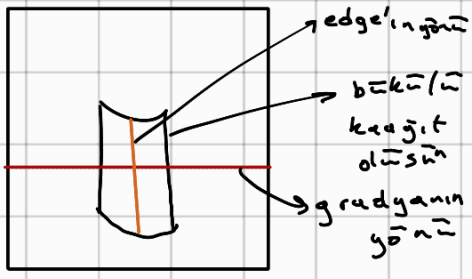
$$G_\sigma(x,y) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{x^2+y^2}{\sigma^2}}$$

2. $M(x,y) = \sqrt{g_x^2(x,y) + g_y^2(x,y)}$

$\theta(x,y) = \tan^{-1}\left(g_y(x,y) / g_x(x,y) \right)$

3- Treshold M: $M_T(x,y) = \begin{cases} M(x,y) & \text{if } M(x,y) > T \\ 0 & \text{O.W.} \end{cases}$

4. Suppress non-maxima pixels in M_t (kalın olmaması için edgelerin)



000 150 160 170 162 142 000

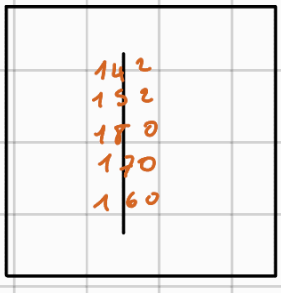
bunlar

5.) İkişer bakıyoruz

z_1 z_2 $z_1 < z_2$

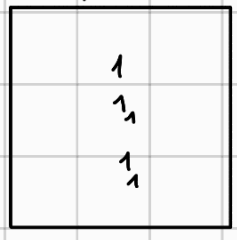
Treshold M_T by z_1 and z_2 to find T_1, T_2

Cannin2.hedefi kırık kenar olmayacak.



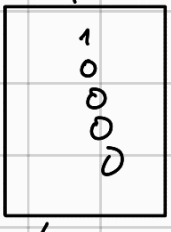
$z_1 = 142$

T_1



$z_2 = 180$

T_2



Bunagöre sola bakiyor varsa devam ediyor